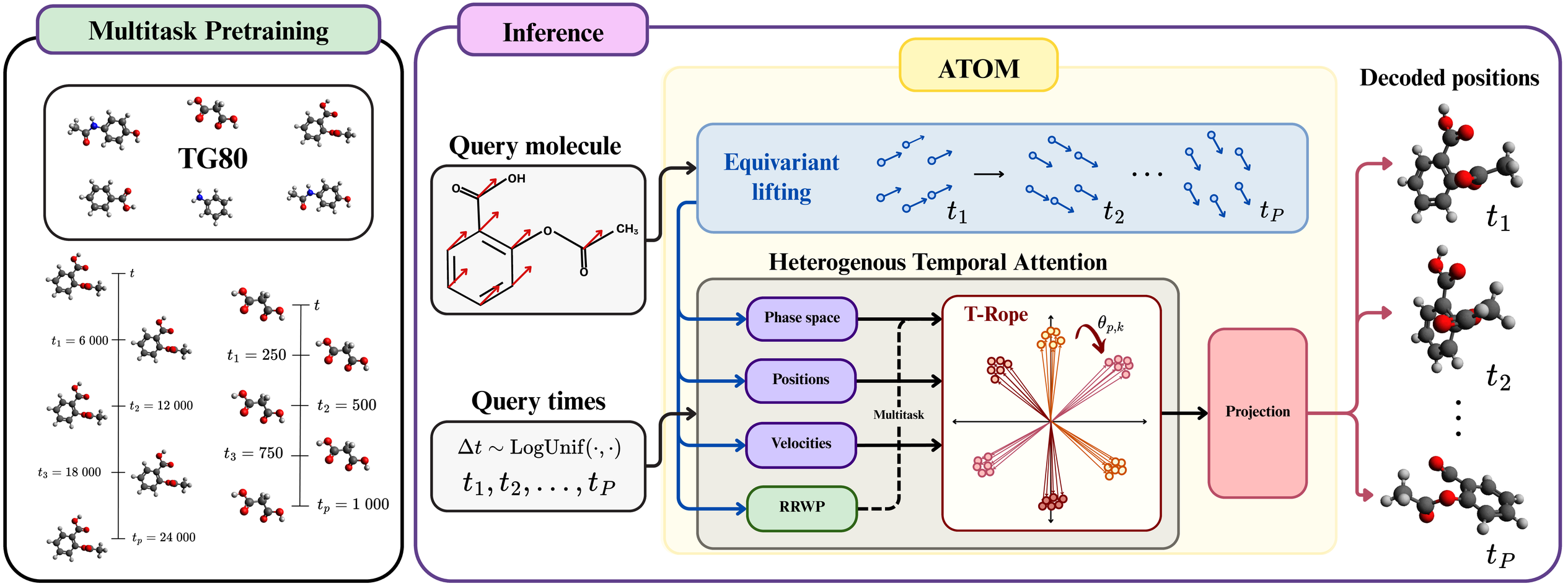

Multitask Pretraining
TG80
t
$t_1 = 6\,000$
$t_2 = 12\,000$
$t_3 = 18\,000$
$t_p = 24\,000$
t
$t_1 = 250$
$t_2 = 500$
$t_3 = 750$
$t_p = 1\,000$
Inference
ATOM
Query molecule
OH
O
O
O
CH_3
Query times
$\Delta t \sim \mathrm{LogUnif}(\cdot,\cdot)$
$t_1, t_2, \ldots, t_P$
Equivariant lifting
t_1
t_2
t_P
Heterogenous Temporal Attention
Phase space
Positions
Velocities
RRWP
Multitask
T-Rope
$\theta_{p,k}$
Projection
Decoded positions
t_1
t_2
t_P